Query Adaptive Transformer and Multiprototype Rectification for Few-Shot Remote Sensing Image Segmentation

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***Abstract*— Deep learning has emerged as a powerful tool for semantic segmentation tasks. However, in some data-deficient and resource-limited scenarios, networks are constrained to learn novel concepts. To tackle this problem, few-shot segmentation (FSS) has been proposed by the machine learning community, aiming at segmenting novel objects by leveraging a handful of annotated samples. However, most advanced FSS algorithms suffer from severe performance degradation when directly applied to remote sensing image (RSI) domains. Challenges arise due to the unique characteristics of RSIs. To address the large intraclass variation brought by various imaging conditions and large-scale variations, a query adaptive transformer (QAT) is proposed. QAT incorporates query priors into the feature extraction process, adapting RSI features to novel objects from query RSIs. It is noteworthy that query priors are extracted via a query prior generation (QPG) module devoid of any query label information. To alleviate interference brought by complex object distribution, a multiprototype strategy is adopted instead of representing objects with a single prototype ambiguously. Moreover, we rectify the prototypes using a prior injection module (PIM), thereby fully leveraging the advantages offered by query priors. The superiority of our method is validated through comprehensive experiments on the public iSAID-5^i dataset and comparisons with state-of-the-art methods. Finally, we propose a novel cross-domain setting to investigate the potential and generalizability of few-shot RSI segmentation for several Earth observation applications.**

***Index Terms*— Cross-domain, data scarcity, few-shot segmentation (FSS), remote sensing image (RSI), Transformer.**

I. Introduction

RAPID evolution of deep learning techniques has led to significant progress in the analysis of remote sensing images (RSIs). Within the domain of Earth observation, automated pixel-wise classification of RSI has emerged as a reliable and scalable approach for monitoring the Earth's surface, facilitating applications such as land cover mapping [1], [2], [3], developmental surveys [4], and disaster damage assessment [5], [6], among others. Despite the demonstrated effectiveness of deep learning frameworks across various Earth observation tasks, their performance deteriorates rapidly in the absence of adequate densely-labeled data or when confronted with unseen classes. Even with additional data, the manual annotation of datasets on a pixel-by-pixel basis is costly and time-intensive, particularly for high-resolution RSIs.

Seeking solutions from the machine learning community, recent progress on few-shot learning [7], [8] make it a promising approach to tackle the problems of data scarcity, which tries to imitate the ability of humans to learn novel concepts quickly. To focus on semantic segmentation tasks more in a low-data regime, the task of few-shot segmentation (FSS) is first proposed [9] and great advances have been achieved. Shaban et al. [9] conducted the pioneering work OSLSM, which designs a classifier weight generation mechanism for each novel class. Based on prototype theory [10] and prototypical networks [7], a prototype-based framework [11] approaches the problem of FSS with metric learning methods. It learns a prototype for each class and measures the cosine similarity between pixels and prototypes to yield the prediction. Wang et al. [12] introduced prototype alignment regularization in order to improve the quality of learned prototypes. Fan et al. [13] proposed a self-support strategy for rectifying prototypes adaptively. While significant advancements have been made, the predominant focus of current research in the field remains on natural image processing, with under-represented exploration into remote sensing applications [14], [15], [16], [17]. As illustrated in Fig. 1, we observe three unique characteristics of the RSI domain.

1) *Various Imaging Conditions:* Diverse imaging conditions such as varying weather conditions and different types of sensors during RSI acquisition contribute to the appearance disparity. For instance, as depicted by the RSIs in the orange area of Fig. 1, distinct levels of illumination and sharpness are evident.
2) *Large Scale Variations:* The scale of geospatial objects varies drastically. Furthermore, even within identical scenes, remote sensing platforms may be positioned at varying elevations, thereby further exacerbating the scale variation of objects in RSIs.
3) *Complex Spatial Distribution:* The spatial distribution of geospatial objects is complex, hindering the localization

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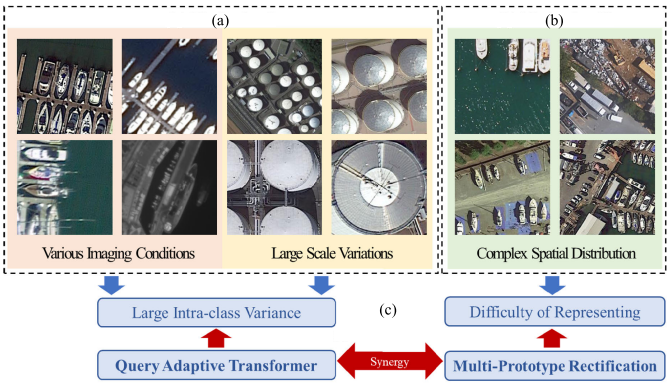


Fig. 1. Unique challenges of few-shot RSI segmentation and our motivations. (a) QAT alleviates the issue of large intraclass variance. (b) Multiprototype rectification is proposed to tackle the difficulty of representation. (c) We aim to propose a mutually reinforcing solution where the two designs synergistically interact to overcome both issues.

and recognition of objects. Images with the same target objects may exhibit significant variability in their backgrounds, as illustrated by the left two RSIs within the green area of Fig. 1. Moreover, the background components within the same image can be distributed complexly, as shown by the right two RSIs in the green area of Fig. 1.

We argue that the aforementioned characteristics contribute to the challenges encountered in few-shot RSI segmentation, primarily falling into two categories: first, **large intraclass variance**, and second, **difficulty in representing**. Moreover, we aim to design the modules in a sequential manner, with the outputs of the preceding module serving as inputs to the next one, thereby forming an integrated and comprehensive approach to the challenges. To fulfill the goal, we propose a pure Transformer-based framework to extract the cues from RSIs and allow them to serve as a kind of prior to guide the segmentation of novel class objects. Specifically, we first extract prior cues from query RSIs with a proposed query prior generation (QPG) module. The extraction process requires no training, thereby ensuring a certain degree of generalizability. To alleviate the issue of large intraclass variance, we then design a query adaptive Transformer (QAT) module to incorporate the prior information while conducting feature extraction of RSIs. In this manner, our model simultaneously learns the relationship between prior cues and RSI features through the self-attention mechanism of transformers [18], thereby adapting the RSI features to be more target-aware. Furthermore, considering the complex spatial distribution of RSIs, we introduce a multiprototype strategy to better represent objects in RSIs. In addition, we employ an attention-based prior injection module (PIM) that leverages the outputted query prior from QAT for prototype rectification, resulting in robust and discriminative multiple prototypes.

In summary, our contributions are threefold.

1) We propose a training-free QPG module to generate prior embeddings seamlessly involved in self-attention calculations of Transformers. The generated prior embeddings encapsulate both the extensive pretraining knowledge and knowledge mined from unlabeled RSI data.
2) We propose two key designs, QAT and PIM, to assist in few-shot RSI segmentation, taking into account the unique properties of RSIs. Specifically, QAT addresses the limitation of large intraclass variation in an adaptive manner and PIM alleviates the interference brought by complex objects in RSIs via prototype rectification.
3) We propose a more realistic cross-domain evaluation setting, examining the generalizability of FSS models while demonstrating the potential applicable scenarios.

II. Related Work

A. Vision Transformer for Earth Observation

Recently, several attempts have been made to leverage the Transformer architecture for capturing global context [19]. Different from convolutional neural network (CNN) structures, the Transformer converts 2-D image-based tasks into 1-D sequence-based tasks. Benefiting from powerful sequence-to-sequence modeling capability, Transformer exhibits superior performance in global information extraction compared to CNN models, achieving cutting-edge performance in natural image vision tasks such as image classification [18], object detection [20], and semantic segmentation [21]. Motivated by this, researchers in Earth observation have implemented versatile transformers for tasks like RSI scene classification [22], [23], object detection [24], change detection [25], image caption [26], [27], RSI super-resolution [28], building and road extraction [29], [30], and with a particular focus on semantic segmentation for RSIs [31], [32], [33], [34], [35].

Specifically, Xu et al. [31] adopt a tiny version of the Swin Transformer for efficient RSI segmentation. Wang et al. [33] proposed a densely connected feature aggregation module with a Transformer backbone. He et al. [34] and Gao et al. [32] combined the Swin Transformer with ResNet to build a hybrid network for parallel feature extraction. Xu et al. [35] made the first attempt to use the Transformer component for RSI land-cover segmentation tasks. All of the aforementioned approaches are based on Transformer encoders, but are implemented with custom-designed CNN decoders. Diverging significantly from traditional CNN approaches, our methodology embraces exclusively Transformer architectures for few-shot RSI segmentation, meticulously tailored to accommodate the distinctive traits of Earth observation datasets.

B. Few-Shot Segmentation

FSS has emerged as a promising research direction aimed at addressing the issue of dense prediction in scenarios with limited data availability. Shaban et al. [9] embarked on pioneering research in the domain of FSS. Their work introduces a novel mechanism for generating classifier weights tailored to individual new classes. Drawing from the principles of prototype theory [10] and the concept of prototypical networks [7], Dong and Xing [11] presented a framework rooted in prototypes, offering an avenue to address the FSS challenge utilizing metric learning techniques. This entails acquiring a prototype representative of each class and yielding predictions

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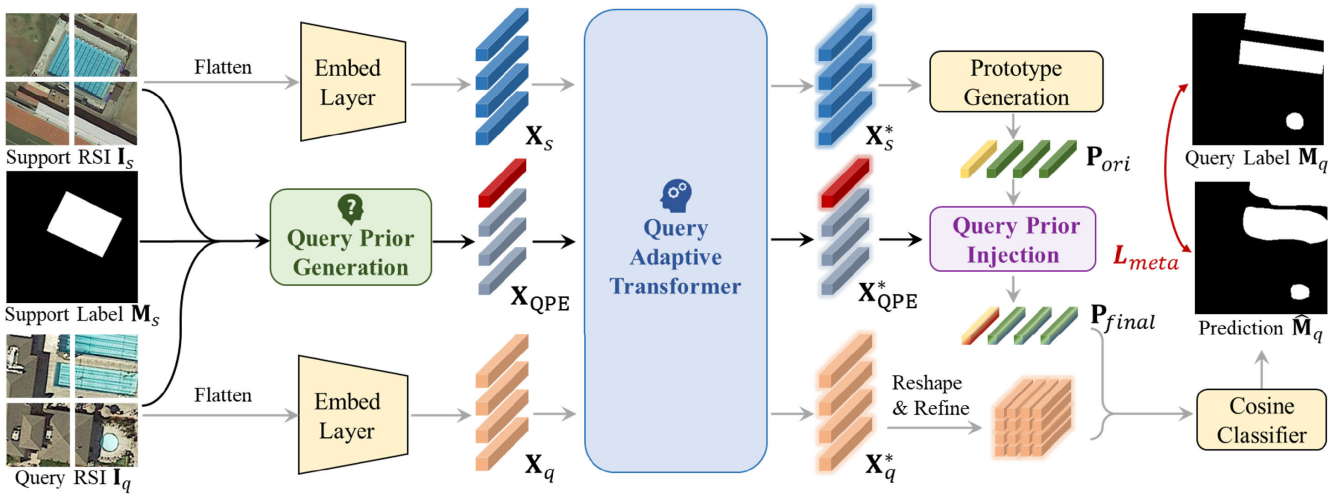


Fig. 2. Overview of the proposed query adaptive network (QANet). The QPG module (see Fig. 3) utilizes query RSI $\mathbf{I}_q$, support RSI $\mathbf{I}_s$, and support label $\mathbf{M}_s$ to generate QPE $\mathbf{X}_{\text{QPE}}$ in a training-free manner. In the subsequent QAT module (see Fig. 4), $\mathbf{X}_{\text{QPE}}$ guides the feature extraction process for both support and query RSIs, from low-level to high-level. Utilizing extracted support RSI features and corresponding labels, a prototype set $\mathbf{P}_{\text{ori}}$ is generated and input into the PIM (see Fig. 5), responsible for rectifying prototypes through adaptive fusion with refined QPE $\mathbf{X}_{\text{QPE}}^*$ (output by QAT). Finally, the query-aware rectified prototypes are employed as classification weights in the cosine similarity classifier, resulting in pixel-by-pixel prediction of the query RSI.

through cosine similarity computation between pixels embeddings and aforementioned prototypes. To enhance the quality of prototypes, Wang et al. [12] introduced prototype alignment regularization. Furthermore, Yang et al. [36] derived multiple prototypes from the proposed Prototype Mixture Models, with each prototype correlating to a part of the image. Fan et al. [13] proposed a self-support strategy for generating high-quality prototypes. Zhang et al. [37] first introduced Transformer architectures into FSS.

Considering the unique characteristics of the remote sensing field, certain endeavors have been undertaken to customize the FSS technique for RSIs. Yao et al. [14] addressed the specific challenges of large-scale variation by introducing the scale-aware focal loss. In [15], a nonparametric metric learning objective is adopted to learn to maximize intraclass similarity while minimizing the interclass similarity. In [16], a pyramid correlation fusion network is proposed to enhance the capacity to compute multiscale pixel-wise semantic correlation in order to mine useful information. Li et al. [38] incorporated SAM [39], a Transformer-based foundational model, to generate over-segmented masks, facilitating guided correlation learning. Wang et al. [40] proposed to enhance query features with support prototypes via cross-attention and also leverage SAM [39] for refined segmentation. Both methods [38], [40] employ Transformer-based models (e.g., SAM) for feature extraction, with the approach in [40] additionally using the Transformer block to aggregate information from support prototypes. In contrast, this article explores simultaneous feature extraction and information aggregation from both the support set and the query set.

Notably, a majority of FSS methods consider a setting where base classes are sampled from the same domain as novel classes. However, Lei et al. [41] observed that when applying these methods to cross-domain scenarios, they fail to generalize due to the large discrepancy in the feature distribution across domains. More practically, a new benchmark is established in [41] to evaluate the cross-domain generalization ability of segmentation models under different domain gaps. In the task of few-shot RSI segmentation, Jiang et al. [15] proposed to train the few-shot learners on natural image datasets as outlined in [9], and subsequently evaluate their performance on RSI datasets. This approach serves to validate the transferability of the methods employed. Given the prohibitive costs or time constraints associated with collecting sufficient in-domain meta-training data for some Earth observation applications, such as disaster response [42], we draw inspiration from [41] to propose a novel cross-domain few-shot RSI segmentation setting. In this context, the domain of the training data for meta-learning differs from that of meta-testing, which enables an investigation into the generalization of meta-knowledge from domains with sufficient training labels to those characterized by limited resources.

III. Proposed Method

A. Problem Setting

In this task, we aim to segment the novel geospatial objects in RSIs with only a few annotated samples. Concretely, the experiment dataset is always split into a training set $\mathcal{D}_{\text{train}}$ and testing set $\mathcal{D}_{\text{test}}$. The labels of the training set $\mathcal{D}_{\text{train}}$ is from categories set $\mathcal{C}_{\text{train}}$ and the labels of the testing set $\mathcal{D}_{\text{test}}$ is from categories set $\mathcal{C}_{\text{test}}$, where $\mathcal{C}_{\text{train}} \cap \mathcal{C}_{\text{test}} = \varnothing$. Following the popular episodic paradigm, both $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ are organized into sets of *episodes*. Each *episode* contains a support set of K-shot $\mathcal{S} = \{(\mathbf{I}_s^{(i)}, \mathbf{M}_s^{(i)}) \mid i \in \{1, 2, \ldots, K\}\}$ and a query set $\mathcal{Q} = \{(\mathbf{I}_q, \mathbf{M}_q)\}$. Here, $\mathbf{I} \in \mathbb{R}^{H_0 \times W_0 \times 3}$ denotes the image with RGB channels and $\mathbf{M} \in \mathbb{R}^{H_0 \times W_0}$ denotes the ground truth binary dense label. Superscript (i) indicates the i-th support sample. Subscripts s and q indicate support and query, respectively. This setup emulates a few-shot scenario, wherein, during each episode, the model is tasked with efficiently assimilating the concepts of new categories from the support

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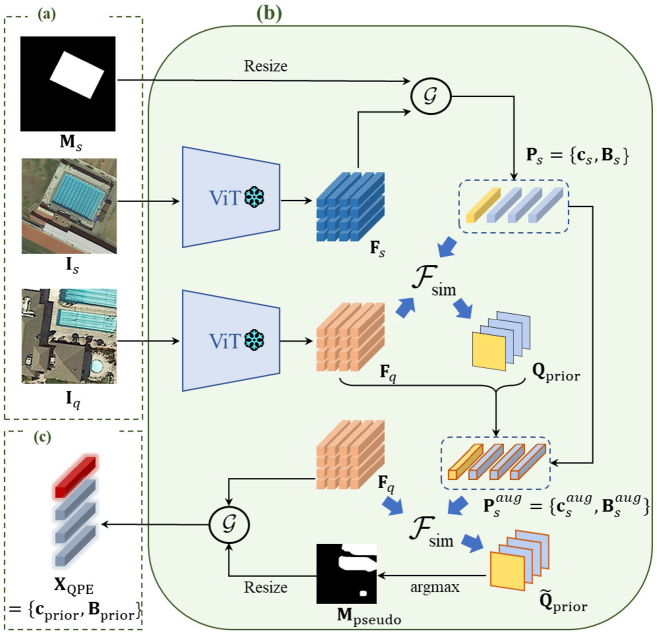


Fig. 3. Illustration of the proposed QPG module. $\mathcal{G}$ denotes the MPG and $\mathcal{F}_{\text{sim}}$ denotes the cosine similarity calculation. (a) Input. (b) QPG. (c) QPE.

set $\mathcal{S}$ and subsequently applying this knowledge to generate predictions for the query image $\mathbf{I}_q$ drawn from the query set $\mathcal{Q}$. The query label $\mathbf{M}_q$ serves solely for evaluating the model's predictions.

B. Method Overview

We propose a novel few-shot RSI segmentation method that achieves effective adaptation to diverse query RSIs. The overall architecture of our proposed network is depicted in Fig. 2, which consists of three key designs: 1) QPG; 2) QAT; and 3) PIM. With the QPG module, a query RSI without a label and several support RSIs with labels are utilized to generate query prior embedding (QPE) in a training-free manner. In the following QAT module, QPE serves as guidance for feature extraction of both support and query RSIs from low-level to high-level, preserving the characteristics of query objects. Based on extracted support RSI features and corresponding labels, we follow the previous method [37], generating multiple prototypes with one representing targets and the others representing different regions of the background. PIM, which we additionally introduced into the pipeline, is responsible for rectifying support prototypes through attention-based adaptive fusion with evolved QPE (output by QAT). Finally, the rectified query-aware prototypes are sent into a cosine similarity classifier [43] as the classification weights with query RSI features, yielding the prediction of the query RSI pixel by pixel. Our model is only supervised by a weighted cross-entropy loss in the meta-training stage but demonstrates promising results.

C. Query Prior Generation

Under the conditions of inadequate training samples (i.e., few-shot scenarios), it is crucial to fully leverage prior knowledge and transfer it to the current task. PFENet [44] transforms high-level features from pretrained ImageNet models into a prior mask indicating the potential presence of targets. The prior generation process is independent of training, thereby ensuring the model's ability to generalize.

In alignment with the spirit of PFENet [44], we introduce a training-free module, referred to as QPG (depicted in Fig. 3). However, our method differs in three key aspects.

1) The proposed QPG module delivers several informative vectors, serving as the query prior instead of one-channel masks in PFENet [44]. We argue that these vectors encapsulate more information from query images and can be seamlessly integrated into the attention mechanism.
2) Given the complexity of RSIs, the prior is generated progressively, ensuring more stable and accurate results.
3) We opt for plain ViTs instead of hierarchical CNNs for compatibility with subsequent modules.

Given a query image $\mathbf{I}_q \in \mathbb{R}^{H_0 \times W_0 \times 3}$ and a support Image $\mathbf{I}_s \in \mathbb{R}^{H_0 \times W_0 \times 3}$ with its mask $\mathbf{M}_s \in \mathbb{R}^{H_0 \times W_0}$, we first extract query features $\mathbf{F}_q \in \mathbb{R}^{H \times W \times M}$ and support features $\mathbf{F}_s \in \mathbb{R}^{H \times W \times M}$ via a frozen ViT [18] Φ_{frozen} pretrained on ImageNet [45]

$$\left[\mathbf{F}_q, \mathbf{F}_s\right] = \mathcal{R}\left(\Phi_{\text{frozen}}\left(\mathbf{I}_q, \mathbf{I}_s\right)\right). \tag{1}$$

Notably, since small-scale geospatial objects are common in RSIs, it is crucial to achieve a fine-scale feature representation from the low-resolution feature map of the original ViTs. Inspired by ViTDet [46], which employs deconvolutions to upsample the spatial dimensions of the ViT feature map by factors of $2\times$ and $4\times$ for finer-scale representations, we empirically observe that a simpler $2\times$ bilinear interpolation, denoted as $\mathcal{R}$, performs effectively as a feature upsampling method in our case, offering a balance between performance and computational cost. Consequently, $H = ((2 \times H_0)/\text{patch_size})$, $W = ((2 \times W_0)/\text{patch_size})$, where patch_size denotes the patch size of ViTs.

Then, we adopt a Voronoi-based partition method [37] for multiple prototype generation (MPG). The Voronoi-based partition method processes $\mathbf{M}_s$ into one foreground mask and N background masks, with each background mask corresponding to a region defined by N Voronoi polygons partitioned from the background of $\mathbf{I}_s$. The MPG (denoted by $\mathcal{G}$) takes these $N+1$ masks and the image feature map as input, producing a prototype set $\mathbf{P}_s$ through common masked average pooling [12], [44]. The prototype set $\mathbf{P}_s = \{\mathbf{c}_s, \mathbf{B}_s\}$ includes one foreground prototype $\mathbf{c}_s \in \mathbb{R}^M$ and a set of N background prototypes $\mathbf{B}_s = \{\mathbf{b}_s^{(i)} \in \mathbb{R}^M \mid i \in \{1, 2, \ldots, N\}$.

Pixel-prototype matching is then conducted between $\mathbf{P}_s$ and the query feature $\mathbf{F}_q$ via a function $\mathcal{F}_{\text{sim}}$, which output the cosine distance between the $N+1$ prototypes and $H \times W$ pixel embeddings

$$\mathbf{P}_s = \mathcal{G}(\mathbf{F}_s, \widetilde{\mathbf{M}}_s) \tag{2}$$

$$\mathbf{Q}_{\text{prior}} = \mathcal{F}_{\text{sim}}(\mathbf{F}_q, \mathbf{P}_s). \tag{3}$$

Here, $\mathbf{Q}_{\text{prior}} = \mathbf{Q}^b_{\text{prior}} \oplus \mathbf{Q}^f_{\text{prior}}$. $\oplus$ denotes concatenation along the channel dimension. $\mathbf{Q}^f_{\text{prior}} \in \mathbb{R}^{H \times W \times 1}$ and $\mathbf{Q}^b_{\text{prior}} \in \mathbb{R}^{H \times W \times N}$ represent the foreground similarity map and N

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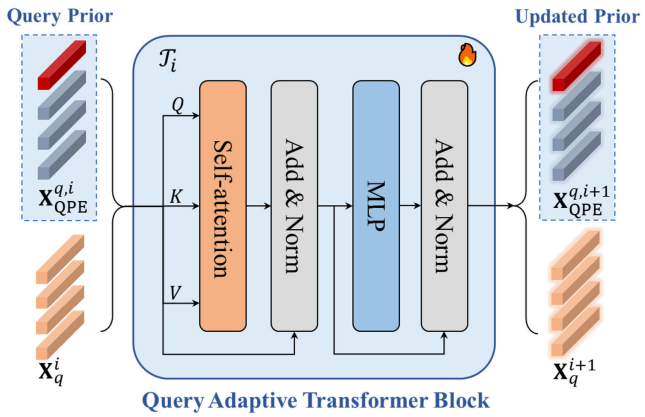


Fig. 4. Illustration of the proposed QAT module. Note that we only visualize the processing of query features; however, the processing of support features follows the same procedure.

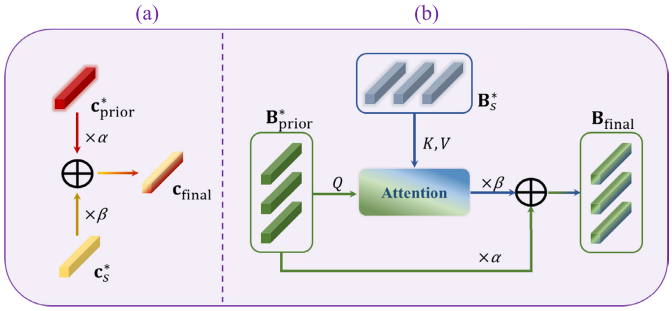


Fig. 5. Illustration of the proposed PIM module. $\oplus$ denotes the element-wise summation. FG and BG denote foreground and background, respectively. (a) FG prior injection. (b) BG prior injection.

background similarity map, respectively, serving as coarse prior masks. $\widetilde{\mathbf{M}}_s \in \mathbb{R}^{H\times W}$ denotes the down-sampled support mask. $\mathcal{G}$ denotes MPG process. Based on the coarse prior from $\mathbf{Q}_{\text{prior}}^f$ and $\mathbf{Q}_{\text{prior}}^b$, we select specific pixel embeddings according to the confidence from $\mathbf{F}_q$ to augment the $\mathbf{P}_s$, in order to bridge the potential appearance gaps between support RSIs and query RSIs

$$\mathbf{c}_s^{\text{aug}} = \mathbf{c}_s + \sum_{x,y} \mathbf{F}_q^{(x,y)} \mathbb{1}\left[\mathbf{Q}_{\text{prior}}^{f,(x,y)} > \tau_f\right] \tag{4}$$

$$\mathbf{b}_s^{\text{aug},(i)} = \mathbf{b}_s^{(i)} + \sum_{x,y} \mathbf{F}_q^{(x,y)} \mathbb{1}\left[\max_n \mathbf{Q}_{\text{prior}}^{b,(x,y,n)} > \tau_b\right]. \tag{5}$$

Leveraging the augmented prototype set $\mathbf{P}_s^{\text{aug}}$, we conduct pixel-prototype matching with $\mathbf{F}_q$ again and obtain finer prior masks $\widetilde{\mathbf{Q}}_{\text{prior}} \in \mathbb{R}^{H\times W\times(N+1)}$, similar to the calculation of $\mathbf{Q}_{\text{prior}}$. The pseudo label $\mathbf{M}_{\text{pseudo}} \in \mathbb{R}^{H\times W}$ of $\mathbf{I}_q$ can be then calculated based on $\widetilde{\mathbf{Q}}_{\text{prior}}$. Furthermore, we condense compact query prior leverage $\mathbf{M}_{\text{pseudo}}$ and the query feature $\mathbf{F}_q$ via another MPG process. The process is formulated as

$$\mathbf{M}_{\text{pseudo}}^{(x,y)} = \begin{cases} 1, & \text{if } \arg\max_{k\in\{0,1,\ldots,N\}} \left(\widetilde{\mathbf{Q}}_{\text{prior}}^{(x,y,k)}\right) = N, \\ 0, & \text{if } \arg\max_{k\in\{0,1,\ldots,N\}} \left(\widetilde{\mathbf{Q}}_{\text{prior}}^{(x,y,k)}\right) < N \end{cases} \tag{6}$$

$$\mathbf{X}_{\text{QPE}} = \mathcal{G}(\mathbf{F}_q, \mathbf{M}_{\text{pseudo}}). \tag{7}$$

Here, $\mathbf{X}_{\text{QPE}}$ is composed of one foreground prior embedding $\mathbf{c}_{\text{prior}} \in \mathbb{R}^M$ and N background prior embeddings $\mathbf{B}_{\text{prior}} = \{\mathbf{b}_{\text{prior}}^{(i)} \in \mathbb{R}^M \mid i \in \{1, 2, \ldots, N\}$. Compared to similarity score maps, QPE captures more high-level information about query images, which enables further interaction in the subsequent QAT design.

D. Query Adaptive Transformer

We propose the QAT as a tool to incorporate prior knowledge reserved in QPE. As illustrated in Fig. 4, compared with naive Transformer blocks, our QAT tackles the pixel embeddings of RSIs (alongside with class tokens) and the additional QPE in a unified manner, updating QPE via self-attention and multilayer perceptron. It is noticed that QPE is also influenced by support RSI and query RSI, respectively, therefore resulting in different versions of QPE. We use a simple average operation to bridge such a gap. The process can be formulated as

$$\begin{cases} \left[\mathbf{X}_{\text{QPE}}^{s,i+1}, \mathbf{X}_s^{i+1}, e_s^{i+1}\right] = \mathcal{T}_i\left(\mathbf{X}_{\text{QPE}}^i, \mathbf{X}_s^i, e_s^i\right) \\ \left[\mathbf{X}_{\text{QPE}}^{q,i+1}, \mathbf{X}_q^{i+1}, e_q^{i+1}\right] = \mathcal{T}_i\left(\mathbf{X}_{\text{QPE}}^i, \mathbf{X}_q^i, e_q^i\right) \end{cases} \tag{8}$$

$$\mathbf{X}_{\text{QPE}}^{i+1} = \frac{\mathbf{X}_{\text{QPE}}^{q,i} + \mathbf{X}_{\text{QPE}}^{s,i}}{2}. \tag{9}$$

$\mathcal{T}_i, i \in \{1, 2, \ldots, 9\}$ denotes the ith Transformer layer of plain ViTs [18]. e_s and e_q denote the default class tokens of support RSI and query RSI as the same in vanilla ViT architectures [18]. Superscript i represents the inputs for the ith Transformer layer. $\mathbf{X}_q \in \mathbb{R}^{HW\times M}$ and $\mathbf{X}_s \in \mathbb{R}^{HW\times M}$ denote image embeddings of $\mathbf{I}_q$ and $\mathbf{I}_s$, respectively, which is generated by embedding function $\mathcal{E}$

$$[e_s, \mathbf{X}_s] = \mathcal{E}(\mathbf{I}_s), \quad [e_q, \mathbf{X}_q] = \mathcal{E}(\mathbf{I}_q). \tag{10}$$

Here, $\mathcal{E}$ denotes the embedding function, which patchifies the whole image and reshapes it into a sequence of patches, transforming them into a series of M-dimensional vectors. The intuition of QANet is that QPE interacts with RSI embeddings via self-attention in every Transformer layer, thus guiding the attention assignment. The flexible modeling capability of self-attention mechanisms can correlate the foreground prior embedding and background prior embedding with corresponding embeddings of RSIs, and then enhance such correlation via modifying both $\mathbf{X}_{\text{QPE}}$ and $\mathbf{X}_s$ (or $\mathbf{X}_q$). Trough experiments, we find that our QAT enables a more proper attention assignment than vanilla ViT [18], where the foreground regions of are more focused and irrelevant attentions are suppressed.

Following the iterative refinement through multiple QAT layers, we obtain the enhanced support RSI feature, refined query RSI feature, and optimized QPE, denoted as $\mathbf{X}_s^*$, $\mathbf{X}_q^*$, and $\mathbf{X}_{\text{QPE}}^*$, respectively. So far we have acquired adequately refined representations of the support and query RSIs, suitable for prototype generation and subsequent prediction tasks.

E. Prior Injection Module

However, as is customary, the conventional practice of deriving prototypes from the support set may not necessarily be suitable for applications in remote sensing. As articulated in our motivation (referring to Fig. 1), the background of RSI

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is complex, and the background of support images may be entirely unrelated to that of the query images. For instance, while the background for ships in the support set may be the ocean, the query images may depict ships grounded on land. Consequently, adjustments to the background prototypes are imperative. In addition, for foreground objects, the foreground in support images may bear resemblance to the background in query images. For instance, in RSIs, certain cars and boats, as well as some containers and buses, exhibit significant similarities, leading to potential confusion. Therefore, rectification for foreground prototypes is also necessary.

We further introduce the PIM, aiming to rectify multi-prototypes derived simply from support RSI features and transform them into query-aware ones (depicted in Fig. 5). Given the output $\mathbf{X}_s^*, \mathbf{X}_q^*$, and $\mathbf{X}_{\text{QPE}}^*$ of the QAT component, the PIM condenses $\mathbf{X}_s^*$ through MPG first, which generates multiprototypes $\mathbf{c}_s^* \in \mathbb{R}^M$ for the foreground and $\mathbf{B}_s^* = \{\mathbf{b}_s^{*,(i)} \in \mathbb{R}^M \mid i \in \{1, 2, \ldots, N\}$ for N different regions of the background. Then, PIM injects the refined QPE $\mathbf{X}_{\text{QPE}}^*$ into these prototypes based on an adaptive weighted aggregation mechanism. As shown in Fig. 5, $\mathbf{c}_{\text{prior}}^*$ denotes foreground query prior, while $\mathbf{B}_{\text{prior}}^*$ denotes background query prior. $\mathbf{c}_s^*$ denotes foreground prototype, while $\mathbf{B}_s^*$ denotes background prototypes. The FG prior injection process is formulated by

$$\mathbf{c}_{\text{final}} = \alpha \cdot \mathbf{c}_{\text{prior}}^* + \beta \cdot \mathbf{c}_s^*. \tag{11}$$

Here, α and β denote the fusion weights of $\mathbf{c}_{\text{prior}}^*$ and $\mathbf{c}_s^*$, respectively. As for the background prior injection, we need to disseminate the information of N background prototypes to the corresponding N query background prior to embeddings. Therefore, it is necessary to unveil the correspondence relationship. Intuitively, we measure the degree of correspondence between $\mathbf{b}_s^{*,(i)}$ and $\mathbf{b}_{\text{prior}}^{*,(j)}$ based on attention mechanism. For simplicity, we stack $\mathbf{B}_s^*$ and $\mathbf{B}_{\text{prior}}^*$ so that they become $N \times M$ matrices. The attention calculation is as follows:

$$Q = \text{Norm}\left(\mathbf{B}_s^*\right), \quad K = \text{Norm}\left(\mathbf{B}_{\text{prior}}^*\right) \tag{12}$$

$$\mathcal{A} = \text{Softmax}(QK^T). \tag{13}$$

Norm represents L2 normalization along the row dimension. The larger the attention score is, the stronger correspondence it exists. We utilize the attention score matrix $\mathcal{A} \in \mathbb{R}^{N \times N}$ as weights to aggregate $\mathbf{b}_{\text{prior}}^{*,(j)}$ adaptively

$$\mathbf{B}_{\text{final}} = \alpha \cdot \mathbf{B}_s^* + \beta \cdot \mathcal{A}\mathbf{B}_{\text{prior}}^*. \tag{14}$$

Here, the sum of tradeoff weights α and β equals 1.0, and α is set to 0.7 according to our ablation study. The final prototypes $\mathbf{P}_{\text{final}} = \{\mathbf{c}_{\text{final}}, \mathbf{B}_{\text{final}}\}$ are utilized to yield final predictions

$$\widehat{\mathbf{M}}_q = \mathcal{F}_{\text{sim}}\left(\mathbf{P}_{\text{final}}, \mathbf{X}_q^*\right). \tag{15}$$

F. Loss Function

Our framework is optimized through a weighted cross-entropy loss $\mathcal{L}_{\text{meta}}$ over the final prediction $\widehat{\mathbf{M}}_q$ and ground truth $\mathbf{M}_q$, as proposed in [47], which emphasizes the segmentation quality of boundary regions. Concretely, a weight map $\mathbf{W}_q$ of the query image is first computed

$$\mathbf{W}_q = \exp\left(\frac{-\mathcal{T}(\neg B)}{\sigma^2}\right) + 1. \tag{16}$$

Here, B represents a binary map derived from $\mathbf{M}_q$, where exclusively the pixels corresponding to the boundaries of objects are assigned a positive value. $\mathcal{T}$ represents the Euclidean distance transform [48], and $\neg$ is the boolean not operation. σ serves as a hyperparameter regulating the decay rate of weights originating from boundary pixels to adjacent pixels. The loss function $\mathcal{L}_{\text{meta}}$ is computed as follows:

$$\mathcal{L}_{\text{meta}} = \frac{1}{H_0 \times W_0} \sum_{i,j} w_{(i,j)} \big(y_{(i,j)} \log \hat{y}_{(i,j)} + \big(1 - y_{(i,j)}\big) \log\big(1 - \hat{y}_{(i,j)}\big)\big) \tag{17}$$

where $w_{(i,j)}$, $y_{(i,j)}$, and $\hat{y}_{(i,j)}$ denote the values located at coordinates (i, j) from $\mathbf{W}_q$, $\mathbf{M}_q$, and $\widehat{\mathbf{M}}_q$, respectively.

G. K-Shot Settings

QANet can be naturally extended to the K-shot setting when $K > 1$. Specifically, we modify the process of the MPG module, which now calculates one foreground prototype and N background prototypes for each image by

$$\left[\mathbf{c}_s^k, \mathbf{B}_s^k\right] = \mathcal{G}\left(\mathbf{F}_s^k, \mathbf{M}_s^k\right), \quad k = \{1, 2, \ldots, K\} \tag{18}$$

where $\mathbf{F}_s^k$ is the feature of kth support RSI, $\mathbf{M}_s^k$ is the corresponding down-sampled dense label. The final foreground prototype $\mathbf{c}_{K-\text{shot}}$ is calculated by taking the average of different $\mathbf{c}_s^k$

$$\mathbf{c}_{K-\text{shot}} = \frac{1}{K} \sum_{k=1}^{K} \mathbf{c}_s^k. \tag{19}$$

We preserve $K \times N$ background prototypes (i.e., $\{\mathbf{B}_s^k | k \in \{1, 2, \ldots, K\}\}$ where $\mathbf{B}_s^k = \{\mathbf{b}_s^{k,(i)} \in \mathbb{R}^M \mid i \in \{1, 2, \ldots, N\}\}$, since backgrounds of RSIs are usually complex and likely to be inhomogeneous. Following such design, the final background prototypes $\mathbf{B}_{K-\text{shot}}$ can be defined as $\{\mathbf{b}_s^{(i)} \in \mathbb{R}^M \mid i \in \{1, 2, \ldots, KN\}$. In the end, we acquire a single foreground prototype and KN background prototypes for the cosine similarity classifier.

IV. Experiments

A. Experimental Setup

In recent years, there has been a discernible upsurge in scholarly investigations concerning FSS. Nevertheless, the practical implementation of FSS techniques within the domain of RSI processing remains notably sparse. For a fair comparison, we follow previous work [14] with their proposed iSAID-5^i dataset. To be specific, iSAID-5^i is a large-scale and finely annotated segmentation dataset, which contains 18 076 RSIs for training and 6363 RSIs for validation with 15 geospatial categories. Under the few-shot setting, these geospatial categories are evenly divided into three folds, while each fold has five classes. The set of class indexes in each fold i is denoted as $5i + x$ where $i \in \{0, 1, 2\}, x \in \{0, 1, 2, 3, 4\}$. To be specific, the novel categories corresponding to each fold are shown in Table I. The few-shot model is tested on a target fold i and trained on the remaining folds. To evaluate the proposed method, we report the mean intersection over

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TABLE I
TESTING CLASSES FOR THE THREEFOLD CROSS-VALIDATION. THE TRAINING CLASSES OF iSAID-5^i, $i = 0, 1, 2$ ARE DISJOINT WITH THE TESTING CLASSES

Split	Novel classes
iSAID-5^0	Ship, Storage Tank, Baseball Diamond, Tennis Court, Basketball Court
iSAID-5^1	Ground Track Field, Bridge, Large Vehicle, Small Vehicle, Helicopter
iSAID-5^2	Swimming Pool, Roundabout, Soccer Ball Field, Plane, Harbor

union (mIoU) of testing classes, following the protocol in [14]. Regarding the selection of novel categories within each fold, we follow the experimental setup of SDM [14], where the categories are determined according to the order of the label dictionary in the original dataset.

In this article, we also propose a novel cross-domain setting, utilizing a landslide extraction dataset [49] and a photovoltaic (PV) panel segmentation dataset [50] for meta-testing. Specifically, for testing landslide extraction tasks, we select all the 770 images with landslide labels from the Bijie Landslide dataset [49] and resize all the images and corresponding dense labels to 256 × 256. For testing PV segmentation tasks, we select the PV01 subset from [50], which comprises 645 UAV aerial images and their corresponding dense labels.

In addition, following [49] and [50], precision, recall, and $F1$ score are reported to quantify the cross-domain performance. All the evaluation metrics adopted in our study are calculated by the following formula:

$$\begin{cases} \text{IoU} = \dfrac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}} + N_{\text{FN}}} \\ F1 = \dfrac{2 \cdot N_{\text{TP}}}{2 \cdot N_{\text{TP}} + N_{\text{FP}} + N_{\text{FN}}} \\ \text{Recall} = \dfrac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \\ \text{Precision} = \dfrac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \end{cases} \tag{20}$$

where N_{TP}, N_{FP}, N_{TN}, and N_{FN} are the pixel number of true positive, false positive, true negative, and false negative, respectively.

B. Implementation Details

In our experiments, all the images are resized to 256 × 256 and augmented following [44]. We evaluate the proposed method using a plain Transformer backbone, ViT-B/16 [18], pretrained on ImageNet-1K [45]. The cross-entropy losses are optimized with boundary-enhanced weight maps introduced by [47]. The proposed model is implemented using the PyTorch framework and all the experiments are conducted on NVIDIA A100 GPUs with Linux operating systems. The average training time for our model across three splits of iSAID-5^i is about 61 min in the one-shot setting, and 155 min in the five-shot setting. We train the model end-to-end using the SGD optimizer, where the weight decay is set to 0.00005 and the momentum is set to 0.9. We choose a constant learning rate of 0.0001 and a batch size of 8. The number of epochs is set to 60 in both one-shot and five-shot settings, with each epoch comprising 2000 randomly sampled episodes.

For the hyperparameters τ_f, τ_b, α, and N of the proposed model, we set them to 0.9, 0.5, 0.7, and 5. σ for weight map calculation is set to 5. A detailed discussion of the hyper-parameter settings is shown in the subsequent ablation study.

C. Performance Analysis

We compare and analyze our model with other advanced FSS algorithms in this section. The superiority of the proposed model is verified from quantitative and qualitative perspectives via careful evaluation. We evaluate our proposed QANet on the challenging iSAID-5^i dataset and compare it against several state-of-the-art FSS methods including generic FSS methods like PMMs [36], PFENet [44], SSP [13], FPTrans [37], and methods specially tailored for RSIs like SDM [14], PCFNet [16], PGSL [40], and MGCL [38]. Experiments are conducted in the one-shot and five-shot settings across different splits of the dataset.

As shown in Table II, our QANet achieves competitive performance. In the one-shot setting, QANet obtains a mIoU of 40.54% across three different splits, surpassing the previous few-shot RSI segmentation method PGSL [40] by 6.37%. Remarkably, in the five-shot scenarios, QANet improves the mIoU metric to 49.51%, exceeding MGCL [38] by 4.18%. The performance comparison presented in Table II highlights the competitive edge of QANet. As depicted in Fig. 6, our QANet yields high-quality predictions in terms of boundary quality and integrity when confronted with novel classes in a one-shot scenario. The strong baseline method, utilizing a ViT-B/16 [18] backbone coupled with a cosine classifier head (identical to the baseline in Tables III and IV), serves as the comparative reference.

We infer that the failure cases are due to the complex internal structures of geospatial objects such as the roundabout (the eighth column in Fig. 6) and the basketball court (the ninth column in Fig. 6), while some components also appear as interference in the query. In addition, qualitative results depicted in Fig. 7 further illustrate the efficacy of our approach under the five-shot settings. We also compare the computation efficiency in terms of frames per second (FPS) in Table V under the five-shot setting and floating point operations per second (FLOPs). While our method has higher computational complexity than CNNs due to the Transformer architecture, it remains computationally efficient among Transformer-based approaches. These results clearly validate the effectiveness of our QANet for FSS.

D. Ablation Studies

We conduct qualitative and quantitative ablation studies to investigate the effectiveness of the proposed designs, including the backbone architectures, the influence on prototype spaces (see Fig. 8) and the choice of crucial hyperparameters. If not specified, all the ablation experiments are conducted on the

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TABLE II

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON iSAID-5^i DATASET IN TERMS OF MIOU (%), FOLLOWING PREVIOUS FEW-SHOT RSI SEGMENTATION WORK [14]. "MEAN" DENOTES THE AVERAGE MIOU SCORES FOR SPLIT-0, SPLIT-1, AND SPLIT-2. † DENOTES OUR REIMPLEMENTATION. THE BEST PERFORMANCE IS HIGHLIGHTED IN BOLD

Method	1-shot				5-shot			
	Split-0	Split-1	Split-2	Mean	Split-0	Split-1	Split-2	Mean
PMMs(ECCV 2020) [36]	19.02	18.51	28.42	21.98	20.89	20.87	31.23	24.33
SDM(TGRS 2021) [14]	34.29	22.25	35.62	30.72	39.88	30.59	45.70	38.72
PFENet(TPAMI 2022) [44]	18.75	17.24	22.09	19.36	19.57	18.43	26.14	21.38
FPTrans(NeuraIPS 2022)† [37]	44.11	25.13	39.91	36.38	53.23	34.75	56.78	48.25
SSP(ECCV 2022)† [13]	39.46	26.36	42.25	36.02	51.70	33.52	49.81	45.01
PCFNet(TGRS 2023) [16]	42.94	27.17	40.06	36.72	52.97	**34.91**	51.62	46.50
PGSL(IGARSS 2024) [40]	40.23	24.92	37.34	34.17	45.83	30.71	47.80	41.45
MGCL(TGRS 2024) [38]	45.21	**30.23**	**47.13**	**40.86**	50.58	31.38	54.02	45.33
QANet(Ours)	**48.03**	28.52	45.06	40.54	**56.37**	34.31	**57.84**	**49.51**

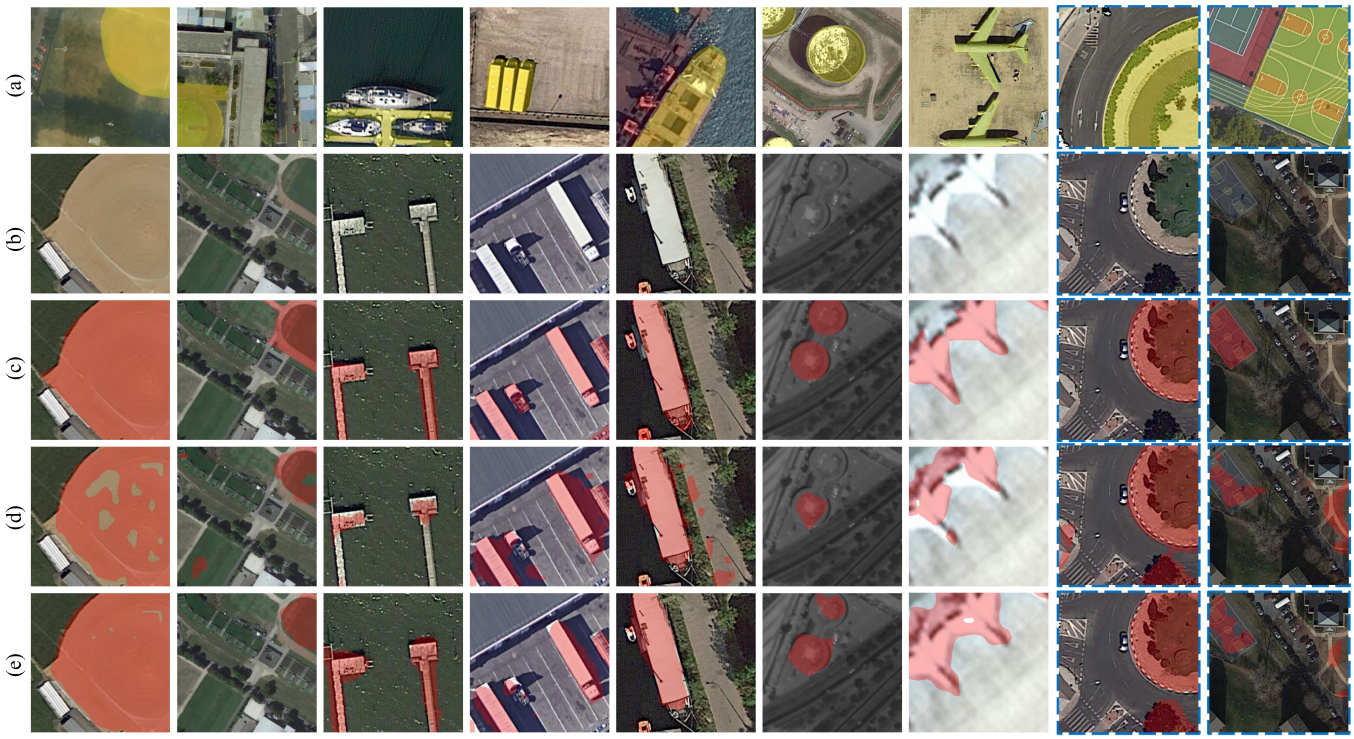


Fig. 6. Qualitative results of the proposed QANet. (Top to bottom) Support images with dense labels (yellow masks), query images, the ground truth of query images (red masks), predictions of our baseline, and predictions of the proposed QANet. Our method can tackle most scenarios where the query differs significantly from the support (as especially shown from the 5th column to the 7th column). Failure cases are highlighted with blue dashed lines. (a) Support. (b) Query. (c) GT. (d) Baseline. (e) Ours.

Split-0 of the iSAID-5^i dataset and the evaluation metric mIoU (%) is reported.

1) Effect of QPG: The proposed QPG module yields condensed prior mined from both labeled support RSIs and unlabeled query RSIs. The QPE facilitates the extraction of target-aware representations in the following QAT module and is leveraged for multiprototype rectification in the PIM. As shown in Table III, without QPG, the performance of our method decreases. We also investigate the choices of foreground confidence threshold τ_f and background confidence threshold τ_b, which are crucial hyperparameters for QPG. As shown in Fig. 9, our QPG performance is more sensitive to changes in τ_f and more robust to changes in τ_b. Overall, as τ_f increases, the performance of our method improves, as higher confidence query features are more likely to belong to target objects, therefore benefiting the augmented prototype set $\mathbf{P}_s^{\text{aug}}$.

2) Effect of QAT: In our method, QAT is proposed to steer our model toward capturing more task-related feature representations, leveraging the QPE generated from QPG. As shown in Fig. 10, we visualize the correlation density of

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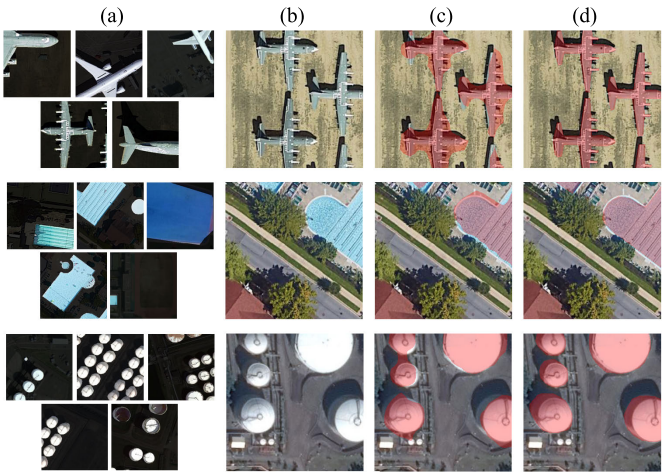


Fig. 7. Qualitative results of proposed QANet under the five-shot settings. The background of support RSIs is overlaid with a translucent black mask, aimed at emphasizing the diversity of targets within the five-shot setting. With more diverse support samples available, QANet yields more precise delineation (i.e., complete and accurate masks). (a) Support. (b) Query. (c) QANet. (d) GT.

TABLE III

ABLATION STUDY OF THE PROPOSED DESIGNS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Module			mIoU(%)	
QPG	QAT	PIM	1-shot	5-shot
✘	✘	✘	43.55	52.32
✘	✘	✔	42.50	50.73
✔	✔	✘	43.54	52.11
✔	✘	✔	46.80	55.18
✔	✔	✔	**48.03**	**56.37**

TABLE IV

ABLATION STUDY OF THE EFFECTS OF THE BACKBONE. ALL THE BACKBONES ARE PRETRAINED ON THE IMAGENET [45] DATASET. † DENOTES OUR REIMPLEMENTATION

Method	Backbone	1-shot	5-shot
SDM [14]	VGG	29.24	36.33
SDM† [14]	RN50	31.52	32.38
SSP† [13]	RN50	39.46	51.70
SDM† [14]	ViT-B/16	8.88	10.24
SSP† [13]	ViT-B/16	44.83	51.68
FPTrans† [37]	ViT-B/16	44.11	53.23
Our baseline	ViT-B/16	43.55	52.32
QANet*(Ours)*	ViT-B/16	48.03	56.37

foreground QPE with pixel embeddings of query RSI and support RSI in the first and last layer of QAT. The red circle highlights the distinction. The correlation map indicates the affinity between the query prior and the corresponding RSI feature in the self-attention computation of QAT. The deeper the red hue at a certain position indicates that the QPE pays greater attention to that location, thus participating more extensively in gathering the corresponding RSI features during QAT computation.

It is observed that the correlation of foreground regions is strengthened after the refinement of multiple QAT layers

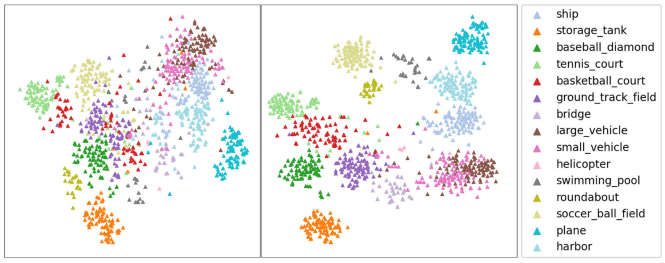


Fig. 8. t-SNE visualization of the foreground prototypes generated (left) without and (right) with PIM. The prototypes are calculated from episodes randomly sampled from iSAID-5^i classes with a maximum of 100 episodes per category for computation.

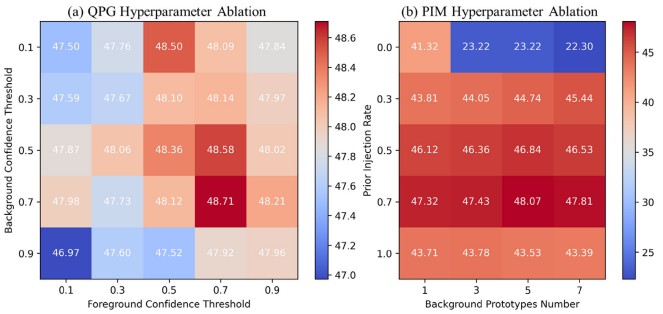


Fig. 9. (a) Ablation studies of foreground confidence threshold τ_f and background confidence threshold τ_b in the proposed QPG. (b) Ablation studies of background prototypes number N and prior injection rate α, which are crucial for our PIM.

TABLE V

COMPARISON OF COMPUTATIONAL COSTS IN TERMS OF FLOPS AND FPS. † DENOTES OUR REIMPLEMENTATION

Method	FLOPs (G)	FPS
SDM† [14]	36.19	28.38
SSP† [13]	68.90	26.68
FPTrans† [37]	337.67	25.97
QANet*(Ours)*	200.74	25.51

(see the third rows in Fig. 10), suggesting that our design optimizes the localization capability of QPE. Notably, certain irrelevant correlations are suppressed, rendering our model more attentive to targets in the query RSI (see the second rows in Fig. 10). An interesting case arises in the first row of Fig. 10, where the support sample comprises various types of boats, some resembling those in the query RSI while others do not. Initially, our model attends to all types of boats in the support sample, yet ultimately, in the last layer, it neglects the boats dissimilar to those in the query RSI (see the red circle), indicating that during the process, regions that are helpful for the prediction of the query are enhanced.

3) Effect of PIM: As depicted in Table III, PIM provides a boost for the performance. Since it fully exploits the benefits brought by the two preceding modules, with QPG providing an initial coarse QPE and QAT refining it by aggregating features relevant to the query object. However, when PIM is not applied, the model fails to leverage QPE, resulting in performance that shows no significant improvement over the

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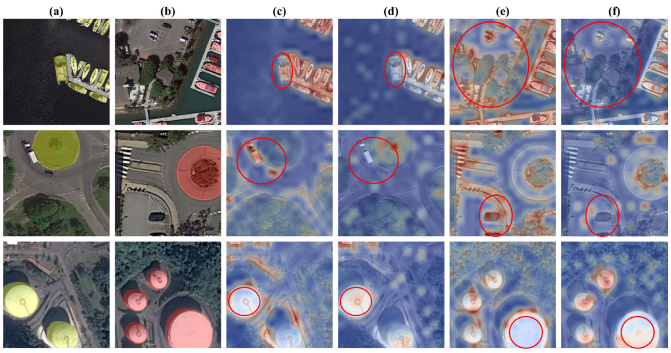


Fig. 10. Visualization of prior correlation intensity in QAT. (a) and (b) Represent the query set and support set. (c)–(f) Represent the correlation maps between foreground QPE and support RSI tokens in the first layer and last layer of QAT, and the correlation maps between foreground QPE and query RSI tokens in the first layer and last layer of QAT, respectively.

baseline. Notably, incorporating only the PIM into our baseline also leads to a slight decline in performance. We attribute it to the exclusion of the QPG and QAT modules, as the support set alone provides information for the prior embeddings. As discussed in Fig. 1, RSIs exhibit significant intraclass variance. Therefore, relying solely on priors from the support set offers limited benefit and may even impair the model's capacity to effectively segment novel targets within the query set.

We conduct a t-SNE visualization of the foreground prototypes generated using QANet with and without PIM. As shown in Fig. 8, PIM significantly helps create more compact and diverse prototype representations (i.e., decreasing intraclass and boosting interclass separability).

We also investigate two key hyperparameters which play a vital role in PIM, as shown in Fig. 9. It is found that a combination of injection rate 0.7 and background prototype number 5 is proper. We infer that query prior benefits the performance within a certain range, and relying on the query prior heavily without incorporating meta-learned knowledge (i.e., $\alpha = 1$) can harm the accuracy.

4) Effect of the Backbone Architectures: In the ablation study, we systematically explore the influence of diverse backbone architectures on the task of FSS. Conceptually, these backbones serve as the architectural foundation of the segmentation model, determining the system's overall robustness and performance. Based on one competitive FSS method for natural image SSP [13] and one advanced FSS method specially designed for RSIs SDM [14], we examine their performance with different backbones. As shown in Table IV, our investigation involves a comparative analysis outlined in the provided table. Each method is associated with a distinct backbone configuration. The method our baseline refers to a plain ViT with a cosine classifier without our QPG, QAT, and PIM modules. In the backbone ablation experiments, all the reimplemented methods are trained for an equal number of rounds, specifically 60 epochs each.

We observe that the performance of SDM [14] improves as the network backbone transitions to more powerful CNNs (i.e., from VGG to RN50). However, when the backbone network switches to ViT-B/16 [18], the performance dramatically declines. We attribute this phenomenon to the fact that SDM is a decoder-based method, where model performance relies on the final convolutional decoder of the network. This decoder is not well-suited to the Transformer-based backbone, leading to a sharp decrease in performance. This conjecture is partially validated by experiments with the SSP [13] method, which shares similarities with our approach. Both methods employ simple linear classifiers (e.g., cosine classifiers), resulting in gains from transitioning to a middle backbone to some extent. Specifically, the one-shot improvement is 5.73, and the five-shot scenario remains largely unchanged. In contrast, our method outperforms the baseline by 4.48 in the one-shot scenario and by 4.05 in the five-shot scenario, surpassing the performance of the similarly Transformer-based method FPTrans [37].

Overall, replacing the CNN architecture with Transformer architecture does not necessarily enhance the performance of previous methods due to compatibility issues with the core designs. Even with improvements, the extent of enhancement remains limited. As exemplified by SSP in Table IV, there is a certain degree of improvement in the one-shot scenario, but the performance decreases instead of improving in the five-shot scenario. Based on the findings, we argue that our QANet prevails owing to the close collaboration between proposed modules and Transformer-based backbones.

TABLE VI

CROSS-DOMAIN EVALUATION ON A LANDSLIDE EXTRACTION DATASET [49] UNDER THE ONE-SHOT AND FIVE-SHOT SETTINGS. "REC" AND "PREC" DENOTE RECALL AND PRECISION, RESPECTIVELY

Method	1-shot				5-shot			
	F1	IoU	Rec	Prec	F1	IoU	Rec	Prec
SDM [14]	25.85	14.85	38.82	19.38	25.69	14.74	37.30	19.59
FPTrans [37]	45.38	29.35	46.39	**44.41**	54.36	37.33	74.90	42.66
QANet(*Ours*)	**47.84**	**31.44**	**54.27**	42.78	**55.75**	**38.65**	**75.78**	**44.09**

TABLE VII

CROSS-DOMAIN EVALUATION ON A PV PANEL SEGMENTATION DATASET [50] UNDER THE ONE-SHOT AND FIVE-SHOT SETTINGS. "REC" AND "PREC" DENOTE RECALL AND PRECISION, RESPECTIVELY

Method	1-shot				5-shot			
	F1	IoU	Rec	Prec	F1	IoU	Rec	Prec
SDM [14]	**73.11**	**57.61**	**86.01**	63.57	69.17	52.88	76.66	63.02
FPTrans [37]	72.27	56.58	66.14	**79.66**	73.78	58.45	66.11	**83.46**
QANet(*Ours*)	72.04	56.30	84.73	62.66	**77.51**	**63.28**	**80.38**	74.83

E. Cross-Domain Setting and Evaluation

For experiments conducted on iSAID-5^i, both meta-training and meta-testing take place within the same domain, rendering the scenario relatively less realistic and posing a diminished challenge for assessing transferability. Therefore, we propose a practical cross-domain setting, in which models are meta-learned on iSAID-5^i [14] and the few-shot capability is tested on a challenging landslide extraction dataset [49] and a PV panel segmentation dataset [50]. Similar to the evaluation protocol of iSAID-5^i, we randomly sample 1000

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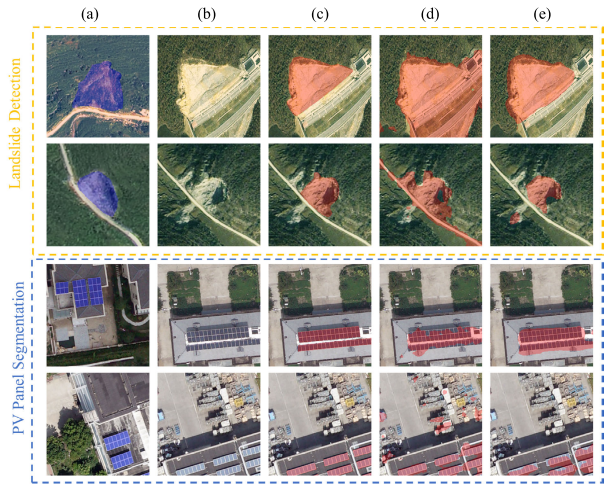


Fig. 11. Qualitative results of our cross-domain evaluation. Our model produces more robust segmentation results compared to SDM on the challenging task of landslide detection (involving irregular geospatial objects) and avoids the interference of complex background objects (as shown in the fourth row). (a) Support. (b) Query. (c) GT. (d) SDM. (e) Ours.

episodes in order to simulate few-shot learning scenarios. These experiments are designed to assess the transferability and potential applicability of the proposed method.

The performance comparison is conducted with an advanced few-shot RSI segmentation method, SDM [14], and a cutting-edge Transformer-based FSS method for natural scenes, FPTrans [37]. Evaluation metrics including IoU, $F1$ score, recall, and precision are reported. As outlined in Tables VI and VII, we report the quantitative results on both tasks. In the majority of scenarios, our approach demonstrates a good level of transferability when the testing data comes from a domain different from that of meta-training. As illustrated in Fig. 11, when compared with SDM, our method eliminates the interference posed by intricate backgrounds (as evidenced in the first and fourth rows, where SDM tends to segment similar background objects). However, we observe that the average performance is worse than the results of meta-testing on iSAID-5^i in terms of IoU for the landslide extraction task, which indicates our practical cross-domain setting poses more challenges to FSS models.

F. Discussions

1) CNN or ViT?: CNN architectures employ a sequence of convolutional and downsampling layers that progressively capture features across different spatial scales. This hierarchical feature extraction is particularly effective for tasks that rely on multiscale information. Conversely, the self-attention mechanism in ViT architectures facilitates long-range dependency modeling during feature extraction, making the resulting features more suitable for tasks that require a comprehensive understanding of global context.

Noteworthy is the observation that Transformer-based methods, under the one-shot setting for PV panel segmentation, exhibit performance comparable to, albeit marginally lower than, that of SDM [14]. We hypothesize this disparity stems from the relatively minor visual discrepancies among PV panels contrasted with significant scale variations. Since plain ViT architectures maintain a single-scale feature map and lack the structural priors inherent in hierarchical CNNs [46], [51], one single support sample is insufficient for the model to capture the multiscale characteristics of novel objects.

Therefore, we believe it is currently not feasible to reach a definitive conclusion regarding the optimal choice for RSI processing. However, the motivation for introducing ViT architectures into our task is twofold as follows.

1) First, our method is built upon the self-attention mechanism of the Transformer architecture, inherently incompatible with CNN-based frameworks.
2) Second, with the advent of the foundation model [52], [53] era, transformers are mostly favored for large-scale pretraining due to their superior scalability. The strong representations of foundation models prove to significantly benefit few-shot learning tasks [54]. Investigating this issue contributes to the democratization of large-scale pretrained models for the Earth observation domain.

2) Potential Applications: We argue that the proposed cross-domain evaluation contributes to demonstrating the potential applications of few-shot RSI segmentation.

1) First, few-shot RSI segmentation facilitates rapid responses by automatically detecting rare geospatial objects, such as landslides (referring to Fig. 11) or damaged buildings, which are often observed during natural disasters and manifest as anomalous phenomena on the Earth's surface. Importantly, the proposed approach does not necessitate any retraining or fine-tuning after meta-learning, and it adeptly learns novel concepts from exceedingly limited samples.
2) Second, RSIs often exhibit significant geographic heterogeneity, leading to a decline in model performance when transferring to regions different from training regions. However, envision a scenario where there arises a need to identify PV panels within a specific geographic region. Leveraging our QANet framework, achieving a relatively robust segmentation performance (as demonstrated in Table VII, with an IoU of 63.28) necessitates annotating merely five samples from the collected imagery. This capability significantly aids in the system's cold start. Subsequently, as the system commences operations, further refinement is achieved through iterative optimization using newly collected data. It is noteworthy that our method typically exhibits superior recall. We posit that this advantage holds particular significance in disaster response scenarios, wherein a certain degree of false alarm rate is deemed acceptable.

V. Conclusion

Considering the complex characteristics of RSIs, we propose a novel and powerful ViT-based framework, which benefits from three tightly coupled modules we designed, namely, QPG, QAT, and PIM. The proposed approach, termed QANet, is comprehensively evaluated on the standard benchmark iSAID-5^i and establishes new state-of-the-art. Moreover,

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our novel cross-domain evaluation not only validates the transferability of the proposed method but also demonstrates the application potentials of FSS for Earth observation. Future efforts could explore leveraging query priors extracted from large language models (LLMs) or large vision-language models (LVLMs) to support the few-shot learning process. Given that text serves as a medium for geospatial knowledge, integrating complementary information inherent in the language modality into FSS tasks might enhance learning efficiency and segmentation performance.

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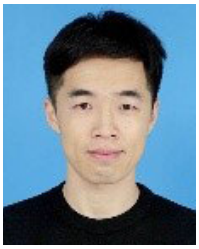
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